



Real-time conflict risk at signalized intersection using drone video: A random parameters logit model with heterogeneity in means and variances

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ABSTRACT

Signalized intersections are crash prone. This can be attributed to driver errors, red light running behaviour, and poor coordination of conflicting traffic. It is anticipated that overall crash risk at signalized intersection would increase when mixed traffic like motorcycles is involved. In this study, a real-time prediction model for motorcycle and non-motorcycle involved conflict risk at the signalized intersection is proposed. For example, high-resolution vehicle and motorcycle trajectory data are extracted from drone videos using advanced computer vision techniques. Additionally, conflict types including rear-end, angle, and head-on conflicts are also considered. Then, the multinomial logit approach is adopted to model the propensity of severe and slight vehicle-vehicle and vehicle-motorcycle conflicts. Furthermore, the problem of unobserved heterogeneity is addressed using the random parameters model with heterogeneity in means and variances. Results indicate that risk of vehicle-vehicle conflict is significantly associated with vehicle speed and acceleration, and conflict type, and that of vehicle-motorcycle conflict is associated with vehicle speed and acceleration, motorcycle lateral speed, conflict type, and time to green signal. Findings should shed light to the development and implementation of optimal traffic signal time plan and traffic management strategy that can mitigate the potential crash risk, especially involving motorcycles, at the signalized intersection.

1. Introduction

1.35 million people are killed on the roads every year worldwide. Road crash is the leading cause of death for children and young adults of age between 5 and 29 years. Unfortunately, 93 % of total road deaths are in developing countries (World Health Organization, 2024). In China, road fatality rate was at a high level at 157 per million population in 2021 (National Bureau of Statistics of China, 2024). Among all road entities, majority of the crashes occur at the intersections. It could be attributed to complicated vehicle maneuvers, repeated stop and go traffic, and frequent conflicts among vehicles of different traffic streams (Wong et al., 2007). Safety studies for intersections based on historical crash data are conducted. Factors including built environment, geometric design, traffic control, environmental and weather conditions, and traffic flow characteristics that affect the crash risk at intersections are identified (Wong et al., 2007; Huang et al., 2008; Wang et al., 2018). However, only the aggregated data on traffic flow characteristics like annual average traffic flow, vehicle mix, and directional split are considered in conventional crash-based safety studies. To better

understand the crash mechanism, and association between crash occurrence and traffic characteristics, it is necessary to collect the data on dynamic traffic flow characteristics. For example, precise data on the location, speed, and acceleration of individual vehicles are available for real-time safety analysis using advanced data acquisition technologies like radar and video camera (Essa and Sayed, 2018; Song et al., 2022a). For example, disaggregated data on traffic volume, queue length, and shock wave speed at the cycle level are available for safety diagnosis of signalized intersections (Essa and Sayed, 2018). Hence, adaptive traffic signal time plan in response to the real-time traffic conditions can be implemented to improve the overall safety and operation efficiency of signalized intersections.

On the other hand, safety studies based on historical crash record also suffer from the problems including under-reporting, imbalanced crash data, and misspecification (Tsui et al., 2009; Chen et al., 2022; Ding et al., 2022). To this end, different surrogate safety measures and crash precursors are proposed for road safety studies (Tarko et al., 2009; Tarko and Lizarazo, 2021). According to the safety pyramid proposed by Hydén (1987), all traffic events can be classified into three categories,

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namely undisturbed passages, traffic conflicts, and road crashes. Traffic conflicts are far more prevalent than road crashes. They can be easily observed and used to proxy the crash risk. It is increasingly popular to adopt traffic conflict for real-time safety assessment. For example, two proximity indicators including post encroachment time (PET) and time to collision (TTC) are most commonly used to measure traffic conflicts and estimate the crash risk (Zheng et al., 2021).

To measure the traffic conflicts, it is necessary to consider the kinematic of different traffic participants including motor vehicles, motorcycles, bicycles, and pedestrians in the mixed traffic environment. In particular, motorcycles, bicycles, and pedestrians are considered vulnerable road users. They are prone to mortality and severe injury in road crashes. Additionally, movement of vulnerable road users are more stochastic, compared to that of motor vehicles. It is more difficult to describe and predict the motion of vulnerable road users. Hence, vehicle-vehicle and vehicle-vulnerable road user conflicts should be estimated separately (Fu et al., 2018; Nasernejad et al., 2023). Likelihood of severe conflicts between vehicle and motorcycle is usually higher since motorcyclists tend to be more risk-taking and aggressive (Song et al., 2022a). Furthermore, studies have also stratified the traffic conflict analysis by collision type such as rear-end and sideswipe. Influencing factors that affect the traffic conflict risk involving different vehicle class and collision type can be identified (Song et al., 2022a). For the association measure between traffic conflict risk and possible influencing factors, econometric methods like multinomial logit and generalized linear regression models are adopted (Jin et al., 2023). To account for the problems including individual heterogeneity and possible correlation, advanced modeling approaches like random parameters model and models with heterogeneity in means and variances are considered (Behnood and Mannering, 2017a,b).

In this study, the main focuses are vehicle-vehicle and vehicle-motorcycle conflicts at the signalized intersection. Factors that contribute to the risk of vehicle-vehicle and vehicle-motorcycle conflicts at the signalized intersection are identified using video observation survey. To model the kinematic of all traffic participants involved in the traffic conflicts, real-time trajectories of vehicles and motorcycles from aerial view will be collected using drone videos. Temporal proximity indicator – post encroachment time (PET) will be adopted. Additionally, collision types including rear-end, angle, and head-on collisions will be considered. Furthermore, random parameters logit model with heterogeneity in means and variances will be developed for the association measure between conflict risk and possible factors. Contribution of this paper is twofold. First, factors of different level in the hierarchical structure (individual versus cycle level) that contribute to the conflicts involving different traffic participants (vehicle-vehicle versus vehicle-motorcycle) will be identified. Second, random parameters logit model with heterogeneity in means and variances will be applied to account for the effects of individual heterogeneity on the occurrence and severity of vehicle-vehicle and vehicle-motorcyclist conflicts. The remainder of this paper is organized as follow. Literature review will be presented in Section 2. Method of data collection and analysis will be given in Section 3. Estimation results and discussions will be provided in Section 4-6. Finally, the concluding remarks will be given in Section 7.

2. Literature review

2.1. Factors affecting intersection safety

Studies have identified the road environment, traffic and driver factors that affect intersection safety based on historical crash data (Abdel-Aty and Wang, 2006; Huang et al., 2008; Wang et al., 2019a; Wong et al., 2007). For the intersection design, factors including number of approaches, junction control, signal time plan, distance between consecutive intersections, road width and curvature, and number of lanes are found affecting the crash frequency at the intersections (Abdel-Aty and Wang, 2006; Wang et al., 2019a). Traffic flow and proportion of

truck in the traffic are positively associated with crash frequency. Additionally, built environment like commercial, industrial and community land use also contribute to the increase in crash risk at the intersections. For driver characteristics, driver age is positively associated with the risk of more severe crash because of the reduction in physical capability of drivers. Furthermore, driving violation behaviour such as red light running and driving under the influence of alcohol and drug also increase the risk of crash at the intersections.

At the signalized intersections, one of the major crash contributory factors is rear-end conflicts among vehicles in the queues (Essa and Sayed, 2018). Factors including volume of approaching traffic, formation of vehicle queues, cycle time, and green time significantly affect the risk of traffic conflicts (Yuan and Abdel-Aty, 2018). In particular, installation of green signal countdown device can mitigate the safety risk at signalized intersections by increasing the drivers' intention to stop at the yellow dilemma zone and deterring the red light running violation in the short term, even that drivers' propensities of risk-taking behaviour may increase in the long run (Islam et al., 2017; Yan et al., 2022). To this end, it is worth exploring the effectiveness of adaptive signal time plan, that is responsive to the real-time traffic volume and speed, in enhancing the overall safety and operation efficiency of signalized intersections (Jin et al., 2021). For example, some empirical studies indicate that adaptive signal time plan significantly reduces the potential crash risk (Khattak et al., 2018; Jin et al., 2021; Lin and Wei, 2023). Nevertheless, association between crash potential, signal time plan, and real-time traffic characteristics should be examined. With the advances in remote sensing and information communication technologies, it is possible to collect the real-time vehicle position and trajectory data using GPS-enabled device on the vehicles like taxis and buses. Studies indicate that dynamic traffic characteristics like average segment speed, sudden acceleration, hard braking, and speeding are positively associated with the crash risk at the intersections (Xie et al., 2013, 2019; Stipanovic et al., 2018; Stipanovic et al., 2019).

On the other hand, safety of vulnerable road users such as motorcyclists, bicyclists and pedestrians has also been a major concern. Unlike safety studies for motor vehicle, there is a lack of comprehensive data for the estimation of crash exposure of vulnerable road users. Precise information on the volume and speed of vulnerable road users at individual road segments and intersections is usually not available (Zhu et al., 2022). Alternate data sources like household travel survey and bike-sharing data are used to estimate the crash exposure at the aggregated level in terms of the number, time and distance of trips involving vulnerable road users for the association measure between possible influencing factors and crash risk involving vulnerable road users (Lee et al., 2015; Sze et al., 2019; Song et al., 2020). The emerging video observation surveys can be conducted to examine the behaviour of vulnerable road users when interacting with the motor vehicles (Zhu et al., 2022; Nasernejad et al., 2023). Microscopic features like jerk (the first derivative of acceleration) profile (Zaki et al., 2014) and motion of body's joints (Zhang et al., 2022) of pedestrians can be extracted. Findings indicate that the speed and acceleration of vehicles and vulnerable road users involved in the interactions can affect the safety risk (Tageldin and Sayed, 2019; Zhu et al., 2022; Song et al., 2022a).

2.2. Safety surrogate measures

Conventional safety studies based on historical crash record have several limitations. First, crash occurrence is rare and random. It can take considerable time period to gather adequate crash cases to establish valid and reliable inferences. Second, police crash report may suffer from the problem of under reporting, missing data, and misspecification of the agencies who are responsible for crash investigation, reporting, and data coding (Tsui et al., 2009; Watson et al., 2015; Ding et al., 2022; Chen et al., 2022; Tarko, 2012; Tarko and Lizarazo, 2021). Furthermore, some dynamic traffic characteristics and key crash contributing factors like vehicle maneuver and driver behaviour are not captured in the

police crash database. In recent years, some researchers have established real-time crash prediction models for freeways and arterial roads, incorporating the traffic parameters like traffic volume, vehicle occupancy, average speed, and vehicle mix (Wang et al., 2017; Yuan and Abdel-Aty, 2018; Bao et al., 2019; Zhang and Abdel-Aty, 2022; Islam and Abdel-Aty, 2023). Additionally, vehicle trajectory data are captured using advanced vehicle sensing, image processing, and crowdsourcing techniques (Yuan and Abdel-Aty, 2018; Wang et al., 2019b) for the association measure between dynamic traffic characteristics and real-time crash risk (Zhang and Abdel-Aty, 2022; Islam and Abdel-Aty, 2023).

To estimate the potential crash risk of different road entities, several surrogate safety measures like time-to-collision (TTC), post encroachment time (PET), and deceleration rate to avoid crash (DRAC) are proposed (Tarko et al., 2009; Tarko 2023). TTC and PET estimate the crash risk based on time proximity for potential collision of vehicles involved. For example, TTC refers to the time remained before two vehicles collide when the velocities of both remain unchanged (Hayward, 1972). On the other hand, PET refers to the difference between the time at which the first vehicle leaving an intersecting zone and that of the second vehicle arriving at that zone (Cooper, 1984). Furthermore, DRAC is a speed-based safety indicator that considers the minimum required deceleration rate for a following vehicle, avoiding the collision with a leading vehicle (Cooper and Ferguson, 1976). Additionally, other dynamic traffic characteristics such as vehicle approaching speed, yielding rate, time to cross or go, and deceleration rate at the intersection are also considered in the traffic conflict risk analysis (Fu et al., 2018; Essa and Sayed, 2018; Patel et al., 2023).

Furthermore, it is quite challenging to model the trajectories of vulnerable road users including pedestrian, bicyclist, and motorcyclist. For example, movement and behaviour of motorcyclist are more chaotic and unpredictable, compared to that of motor vehicles (Song et al., 2022a; Ospina-Mateus et al., 2023). Factors including road geometry, lane configuration, traffic mix, weaving movement, and variation in speed among all vehicles significantly affect the risk of traffic conflicts involving vulnerable road users. Additionally, sudden acceleration, hard brake, and shockwave propagation are all positively associated with the conflict risk (Sun et al., 2020; Ospina-Mateus et al., 2023; Song et al., 2022a; Nasernejad et al., 2023). Nevertheless, there are very few studies that focus on the modeling of traffic conflict risk involving motorcycle at the signalized intersections, where the interactions among motor vehicles and motorcycles are sophisticated.

2.3. Analytic methods

Because of the discrete nature of crash and conflict severity, discrete outcome models such as binary and multinomial logit or probit regression, and ordered logit or probit regression are adopted to estimate the influences of possible risk factors (Xin et al., 2017; Zhai et al., 2019; Song et al., 2020). Accounting for the effect of spatial dependence on the association measure, advanced approaches like conditional autoregressive and Bayesian spatial method are also adopted (Zeng et al., 2019; Guo et al., 2019). Furthermore, individual heterogeneity attributed to some unobserved and unmeasurable factors like driver attitude, safety perception and driving style can also modify the association between crash risk and other observed factors. To this end, random parameters, random effects, and latent class methods are adopted, accounting for the effect of unobserved heterogeneity in the crash and conflict severity models (Behnood and Mannering, 2017a,b; Fountas et al., 2018; Song et al., 2020; Chang et al., 2021; Sun et al., 2023; Goswamy et al., 2023). Furthermore, advanced approaches like heterogeneity in means and variances and correlated grouped random parameters methods are adopted for the modeling of lane changing behaviour and traffic conflict risk at the highway merging and diverging areas, based on the trajectories of individual vehicles (Li et al., 2021; Song et al., 2022a). Some recent studies accounting for unobserved heterogeneity in the crash and conflict severity models are summarized

in Table 1. Even that models accounting for individual heterogeneity in traffic conflict analysis are prevalent, very few visit the conflicts involving vulnerable road users at the intersections.

3. Data

In this study, a comprehensive high definition drone-based vehicle trajectory dataset – Signalized Intersection Dataset (SIND) is adopted for traffic conflict analysis, both involving and not involving motorcycles, at the intersection (Xu et al., 2022). Overall, a total of seven-hour long videos from 15 days in July and August 2021 were captured. Selected site is a typical four-leg signalized intersection in Tianjin City of China. There are high traffic flow and traffic mix during the observation period. For example, the average number of vehicles approaching the intersection is 700 per hour, and that of motorcycles is 470 per hour respectively. Among the vehicles, majority are private cars (96.8 %). All surveys were conducted in the daytime. The dimensions of the area covered by the drone video are 80×50 m.

Table 1

Recent road safety studies with econometric models considering unobserved heterogeneity.

Study	Econometric model	Crash type	Location (Study period)
Behnood and Mannering (2017a,b)	Random parameters logit model with heterogeneity in means and variances	Vehicle-bicycle crash	Los Angeles, United States (2010–2016)
Fountas et al. (2018)	Grouped random parameters binary logit model	Highway crash	Washington State, United States (2011–2013)
Song et al. (2020)	Hierarchical Bayesian random effects logit model	Pedestrian crash	North Carolina, United States (2007–2018)
Yu et al. (2020)	Random parameters logit model with heterogeneity in means and variances	Rear-end crash at work zone	North Carolina, United States (2010–2013)
Chang et al. (2021)	Latent class clustering and latent segmentation random parameters ordered logit model	Vehicle-motorcycle crash	Queensland, Australia (2012–2016)
Li et al. (2021)	Random parameters logit model with heterogeneity in means and variances	Conflict at highway merging area	Los Angeles, United States (2005–2006) ¹
Song et al. (2022a)	Grouped random parameters logit model with heterogeneity in means and variances	Conflict at highway toll plaza	Hong Kong (2022)
Alnawmasi and Mannering (2022)	Random parameters negative binomial and random parameter logit model with heterogeneity in the means and variances	Highway crash	Kansas, United States (2009–2013)
Jin et al. (2023)	Random parameters logit model with heterogeneity in means and variances	Tunnel crash	California, United States (2018–2020)
Goswamy et al. (2023)	Random parameters ordered logit model with heterogeneity in means	Pedestrian crash at intersection	Florida, United States (2012–2019)
Sun et al. (2023)	Random parameters logit model with heterogeneity in means	Intersection crash	Shenyang City, China (2015–2017)
Gao and Zhang (2024)	Random parameters logit model with heterogeneity in means and variances	Vehicle and e-bike crash at intersection	United Kingdom (2020–2022)

Note:

¹ Next Generation Simulation (NGSIM) Open Data.

Fig. 1 and Fig. 2 illustrate the conflicts involving two vehicles. As shown in Fig. 1, conflicts are stratified into three types by intersecting angle, namely rear-end, head-on, and angled conflicts (Wu et al., 2020; Arun et al., 2021).¹ As shown in Fig. 2, “Vehicle 1” denotes the leading vehicle, and “Vehicle 2” denotes the following vehicle respectively in an interaction. In a rear-end conflict (as shown in Fig. 2(a)), vehicle in the front is the leading vehicle. In an angled or head-on conflict (as shown in Fig. 2(b)), vehicle that reach the intersecting area first is the leading vehicle. In vehicle-motorcycle conflicts, a motorcycle can either be the leading or following vehicle.

In this study, severity levels of traffic conflict are also considered. For example, threshold values for slight traffic conflict and severe conflict are set at post encroachment time (PET) of 4 s and 2 s respectively (Zheng et al., 2019; Patel et al., 2023). Fig. 3 illustrates the heatmaps of traffic conflict density, by vehicle class and severity level, at the study area. As shown in Fig. 3(a), slight vehicle-vehicle conflicts are prevalent at the center of intersection and intersection approaches. As shown in Fig. 3(c) and Fig. 3(d), both slight and severe vehicle-motorcycle conflicts are prevalent at the center of intersection only. Fig. 4 presents the signal time plan of the study area. In particular, average cycle time is 67 s, average green time for major road (Phase 1) is 33 s, and average green time for minor road (Phase 2) is 26 s respectively. Additionally, permissive left-turn is adopted for both Phase 1 and Phase 2. Hence, left-turning traffic have to give way to oncoming traffic. This may induce potential conflicts between left-turning and oncoming (straight-through) traffic (Song et al., 2022b). Table 2 summarizes the data used in this study. Overall, trajectories of 3,545 vehicle-vehicle and 2,601 vehicle-motorcycle interactions are used for the estimation. For each interaction, parameters like longitudinal speed, acceleration rate, and deceleration rate of every vehicle involved are estimated. Additionally, time to green (time remaining before green signal starts) is also considered. Furthermore, for each signal cycle, data including traffic volumes by direction, cycle time, green time for major road, and green time for minor road are captured.

4. Method

In this study, risk of traffic conflict at the intersection, with and without the involvement of motorcycle, will be estimated. Given the discrete nature of the occurrence and severity level of traffic conflict, multinomial logit regression approach would be adopted (Figueroa-Medina, 2005; Behnood and Mannering, 2017a, 2017b).

The utility function of logit model is given by,

$$s_{kn} = \beta_k x_{kn} + \varepsilon_{kn} \quad (1)$$

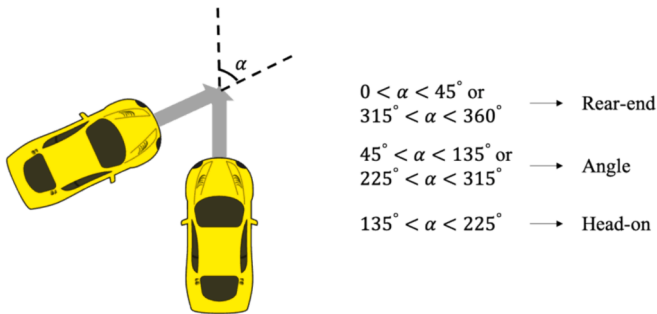


Fig. 1. Conflict type by intersecting angle.

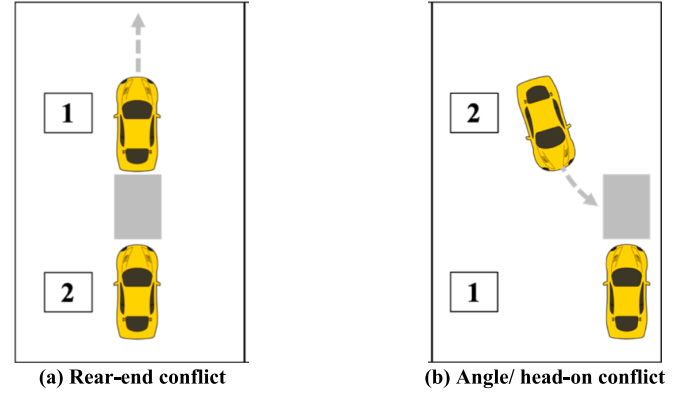


Fig. 2. Vehicle notation.

where s_{kn} is a function determining the probability of conflict outcome k in the observation n , x_{kn} is a vector of explanatory variables that affect conflict outcome, β_k is a vector of estimable parameters, and ε_{kn} is the error term, assuming independent and identically distributed (IID) Type 1 extreme value distribution.

Accounting for the effect of unobserved heterogeneity in the means and variances of random parameter, let β_{kn} be a vector varying across the observations, as defined in Seraneeprakarn et al. (2017),

$$\beta_{kn} = \beta + \Theta_{kn} z_{kn} + \Gamma \omega_{kn} \quad (2)$$

where β is the mean parameter estimated across all observations, z_{kn} is a vector of explanatory variables that capture heterogeneity effect, Θ_{kn} is a vector of estimated parameters for heterogeneity effect, ω_{kn} is normally distributed with $N(0, \sigma_{kn}^2)$, Γ is the symmetric Cholesky matrix based on Cholesky decomposition.

With observation-specific unobserved heterogeneity, the vector β_{kn} have a continuous density function $Prob(\beta_n = \beta) = f(\beta|\varphi)$, and φ is a vector of parameters characterizing this function. Thus, the probability of observation n having conflict outcome k can be given by,

$$p_n(k) = \int \frac{EXP[\beta_k x_{kn}]}{\sum_{k=1}^K EXP[\beta_k x_{kn}]} f(\beta|\varphi) d\beta \quad (3)$$

Additionally, the variance-covariate matrix can be written as,

$$C = \Gamma \Gamma^T \quad (4)$$

with which the diagonal elements are the standard deviation of random parameters, and off-diagonal elements are the variance between random parameters, respectively.

Maximum likelihood approach is adopted for parameter estimation (Washington et al., 2011). Hence, the log likelihood function is given by,

$$LL(\beta_{kn}) = \sum_{k=1}^K \sum_{n=1}^N y_{kn} \frac{1}{R} \sum_{r=1}^R P_{kn}(x_k | \beta_{kn}, \nu_{kn}^r) \quad (5)$$

where r is number of Halton draws.

In this study, 1000 Halton draws are adopted for the convergence of parameters, trading-off between model performance and computation time (Song et al., 2022a). Nevertheless, different function forms including uniform, lognormal, triangular, and normal distributions are considered for the random parameters.

5. Results

5.1. Vehicle-vehicle conflict

Table 3 illustrates the results of goodness of fit assessment of the candidate models. For example, performance metrics like Akaike's

¹ Some studies also analyze the risk of sideswipe conflict. In this study, number of sideswipe conflict is negligible. Hence, sideswipe conflict is not considered.

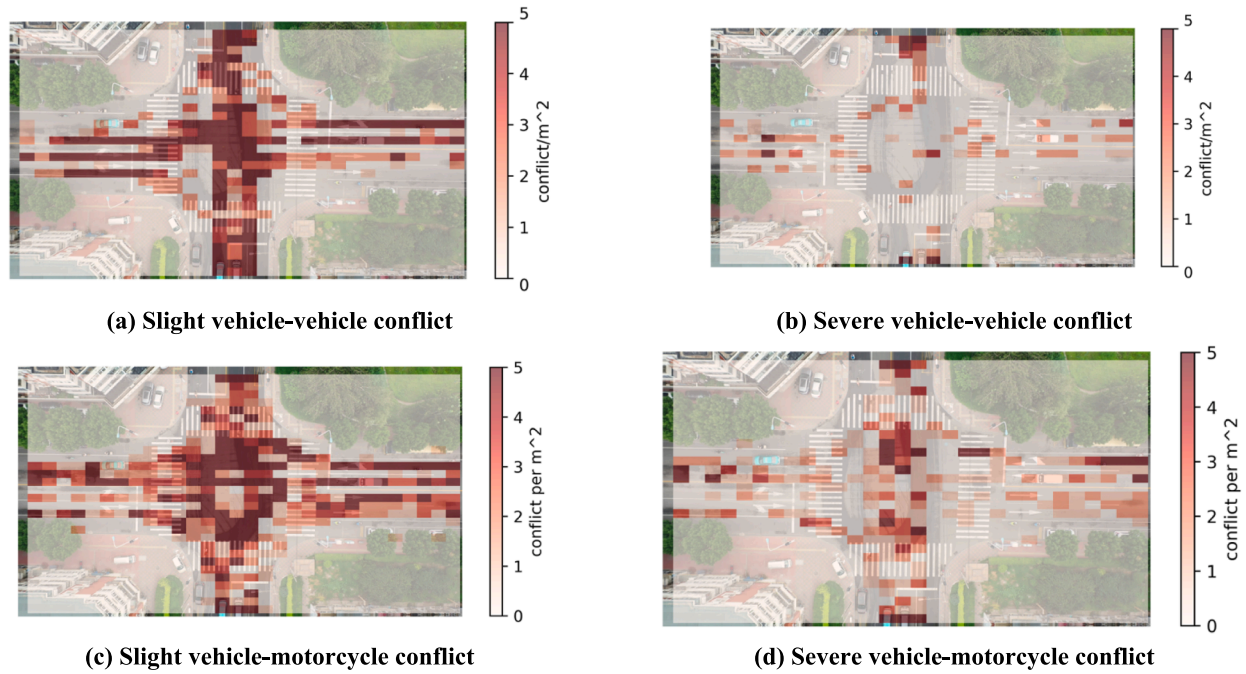


Fig. 3. Traffic conflict heatmap.

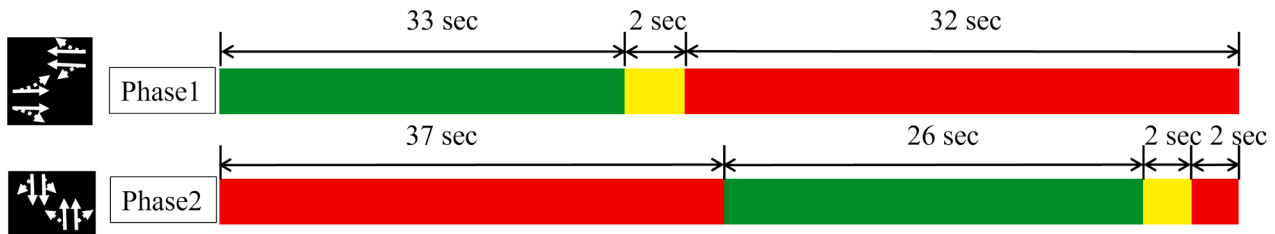


Fig. 4. Signal time plan.

information criterion (AIC) and Bayesian information criterion (BIC) are adopted, considering the model complexity. Smaller AIC and BIC values indicate superior model fit. As shown in Table 3, the random parameters logit model with heterogeneity in means and variances is considered, given the lowest AIC values among the candidate models.

Table 4 presents the results of parameter estimation of random parameters logit model with heterogeneity in means and variances for vehicle-vehicle conflict. As shown in Table 4, rear-end conflict has a higher likelihood of being severe (1.664), compared to other conflict types, at the 1 % level of significance. Additionally, longitudinal speed (slight conflict: 0.106; severe conflict: 0.174) and lateral speed (slight conflict: 2.236; severe conflict: 4.349) of the leading vehicle are positively associated with the likelihood of slight and severe conflicts, all at the 1 % level of significance. On the other hand, longitudinal speed of the following vehicle (slight conflict: 0.093; severe conflict: 0.178) is positively associated with the likelihood of slight and severe conflicts at the 1 % level of significance. Furthermore, the longitudinal acceleration of the leading vehicle is negatively associated with the likelihood of slight conflict (−1.364), but positively associated with that of severe conflict (0.237), both at the 1 % level of significance. Longitudinal acceleration of the following vehicle is negatively associated with the likelihood of severe conflict (−0.265) at the 10 % level. Last but not least, heterogeneity in means and variances of the random parameters are also considered. For the likelihood of slight conflict, both mean and variance of the random parameters of longitudinal acceleration are higher for rear-end conflict, compared to other conflict types, both at the

1 % level of significance.

5.2. Vehicle-motorcycle conflict

Table 5 illustrates the results of goodness of fit assessment of the candidate models. For instance, the random parameters logit model with heterogeneity in means and variances is considered, given the lower AIC and BIC values among the candidate models.

Table 6 presents the results of parameter estimation. For example, time to green (0.631) is positively associated with the likelihood of severe conflict, at the 1 % level of significance. Additionally, longitudinal speed (slight conflict: 0.540; severe conflict: 1.202), lateral speed (slight conflict: 0.586; severe conflict: 0.832), and longitudinal acceleration (slight conflict: 2.095; severe conflict: 3.915) of vehicle are positively associated with the likelihood of slight and severe conflict, all at the 1 % level of significance. Furthermore, rear-end conflict has a higher likelihood of being severe (1.132) at the 1 % level of significance. However, lateral speed of motorcycle is negatively associated with the likelihood of severe conflict (−2.111) at the 1 % level of significance. Last but not least, for the likelihood of severe conflict, vehicle lateral speed is positively associated with the mean of the random parameters of motorcycle lateral speed at the 1 % level of significance.

6. Discussions

In this study, roles of possible factors, both at the individual (e.g.,

Table 2
Summary statistics.

Scope of work	Variable	Vehicle-vehicle conflict				Vehicle-motorcycle conflict			
		Min	Max	Mean	S.D.	Min	Max	Mean	S.D.
Dependent variable									
Conflict severity	No conflict	0	1	0.53	0.58	0	1	0.55	0.50
	Slight conflict	0	1	0.41	0.49	0	1	0.28	0.45
	Severe conflict	0	1	0.06	0.24	0	1	0.17	0.38
Individual-level factor									
Signal time	Time to green (sec)	0	69.95	33.27	19.54	0.03	69.94	32.98	19.55
Leading vehicle characteristics	Longitudinal speed (m/s)	0	9.32	4.79	2.38	0	8.36	3.55	2.55
	Lateral speed (m/s)	0	0.34	0.07	0.11	0	0.23	0.05	0.07
	Longitudinal acceleration (m/s ²)	0	0.86	0.67	1.03	0	0.87	0.61	0.77
	Longitudinal deceleration (m/s ²)	0	0.89	0.18	0.37	0	0.2	0.18	0.34
Following vehicle or motorcycle characteristics	Longitudinal speed (m/s)	0	9.66	4.18	3.19	0	7.74	6.17	2.06
	Lateral speed (m/s)	0	0.36	0.05	0.11	0	0.23	0.05	0.07
	Longitudinal acceleration (m/s ²)	0	0.60	0.36	0.55	0	0.69	0.50	0.74
	Longitudinal deceleration (m/s ²)	0	0.82	0.08	0.24	0	0.51	0.06	0.14
Conflict type	Rear-end conflict	0	1	0.72	0.45	0	1	0.52	0.49
	Angle conflict	0	1	0.16	0.37	0	1	0.34	0.47
	Head-on conflict	0	1	0.11	0.32	0	1	0.14	0.35
Group-level factor									
Signal time	Cycle length (sec)	59.93	72.00	67.09	4.54	59.93	71.74	67.25	4.43
	Green time – major road (sec)	25.93	36.06	32.78	4.45	25.92	36.07	32.99	4.38
	Green time – minor road (sec)	25.49	31.03	26.47	1.45	25.49	31.03	26.43	1.41
Traffic flow characteristics	Left-turn traffic per cycle (veh)	13	33	21.15	5.31	13	33	21.15	5.31
	Right-turn traffic per cycle (veh)	15	36	22.73	5.21	15	36	22.73	5.21
	Straight-through traffic per cycle (veh)	33	64	48.69	8.94	33	64	48.69	8.94

Table 3
Model performance.

Parameter	Random parameters logit model with heterogeneity in means and variances	Random parameters logit model
Loglikelihood	−2944.144	−3007.798
McFadden R-squared	0.244	0.230
AIC	5918.400	6039.700
BIC	6010.900	6113.700

speed and acceleration of vehicle and motorcycle) and cycle (e.g., cycle time and green time) levels, in the risk of vehicle-vehicle and vehicle-motorcycle conflicts at the signalized intersections are examined, based on the real-time trajectory data collected using drone. Marginal effects of the influencing factors on the risk of vehicle-vehicle and vehicle-motorcycle conflicts are summarized in Table 7.

6.1. Vehicle and motorcycle attributes

For the vehicle attributes, one unit increase in the longitudinal speed of the leading vehicle is associated with 8.0 % and 4.8 % increases in the risk of slight and severe vehicle-vehicle conflicts. Similarly, one unit increase in vehicle longitudinal speed is associated with 5.4 % and 7.0 % increases in the risk of slight and severe vehicle-motorcycle conflicts. Such findings are consistent with that of previous studies (Stipancic et al., 2019). For example, Stipancic et al. (2019) found that average speed and hard braking were positively associated with the frequency of severe crashes. Moreover, vehicle speed plays an important role in the injury outcome for crashes involving vulnerable road users, e.g. motorcyclists (Wu et al., 2018; Arun et al., 2021). To this end, advanced traffic control measures like flashing signs and automated speed enforcement cameras could be installed to reduce the speed of approaching traffic (Li et al., 2021; Thomas et al., 2008; Zhang et al., 2022). Furthermore, one unit increase in the longitudinal speed of the following vehicle is associated with 6.6 % and 4.3 % increases in the risk of slight and severe vehicle-vehicle conflicts. They are comparable to

Table 4
Results of parameter estimation for vehicle-vehicle conflict.

Variable	Slight conflict		Severe conflict	
	Coefficient	t-statistic	Coefficient	t-statistic
Intercept	−0.967**	−6.370	−5.300**	−15.220
Longitudinal speed of leading vehicle	0.106**	5.520	0.174**	4.910
Longitudinal speed of following vehicle	0.093**	5.540	0.178**	6.750
Lateral speed of leading vehicle	2.236**	4.420	4.349**	6.280
Longitudinal acceleration of leading vehicle	Mean	−1.364**	−3.430	0.237**
	S.D.	2.434*	2.050	3.670
Longitudinal acceleration of following vehicle	—	—	−0.265*	−1.760
Rear-end conflict	—	—	1.664**	7.880
Heterogeneity in means: Normal distribution				
Longitudinal acceleration of leading vehicle: Rear-end conflict	3.697**	3.620	—	—
Heterogeneity in variances				
Longitudinal acceleration of leading vehicle: Rear-end conflict	1.172**	1.750	—	—

*Marginal at the 10% level.

* Significant at the 5% level.

** Significant at the 1% level.

— Not included in the model.

that in previous studies (Ma et al., 2018; Song et al., 2022a). Indeed, rear-end conflict has been the dominating conflict type at the intersection. Other than automated enforcement system, advanced traffic management measures like signal count down device can reduce the anxiety of driver when approaching the intersection, and therefore, the

Table 5
Model performance.

Parameter	Random parameters logit model with heterogeneity in means and variances	Random parameters logit model
Log likelihood	−2469.893	−2476.444
McFadden R-squared	0.136	0.130
AIC	4961.900	4974.000
BIC	5049.900	5039.400

Table 6
Results of parameter estimation for vehicle-motorcycle conflict.

Variable	Slight conflict		Severe conflict	
	Coefficient	t-statistic	Coefficient	t-statistic
Intercept	0.267**	3.460	−2.147**	−10.670
Vehicle longitudinal speed	0.540**	3.510	1.202**	5.350
Vehicle lateral speed	0.586**	3.270	0.832**	3.160
Vehicle longitudinal acceleration	2.095**	3.230	3.915**	5.060
Motorcycle lateral speed	Mean	—	−2.111**	−2.580
	S.D.	—	2.239**	2.440
Rear-end conflict	—	—	1.132**	7.870
Time to green	—	—	0.631**	2.840
Heterogeneity in means (normal distribution)				
Motorcycle lateral speed:	—	—	2.174**	2.660
Vehicle lateral speed	—	—	—	—
Heterogeneity in variances				
Motorcycle lateral speed:	—	—	−0.048*	−0.110
Time to green	—	—	—	—

*Significant at the 10% level.

* Significant at the 5% level.

** Significant at the 1% level.

— Not included in the model.

potential rear-end conflicts between pair of vehicles (Tarko and Lizarazo, 2021).

On the other hand, one unit increase in lateral speed is associated with 2.4 % and 1.9 % increases in the risk of slight and severe vehicle-vehicle conflicts. Similarly, one unit increase in vehicle lateral speed is associated with 3.0 % and 2.6 % increases in the risk of slight and severe vehicle-motorcycle conflicts. Lateral speed is closely correlated to the lane changing intention of drivers (Das et al., 2020). To mitigate the

potential crash risk, special road markings can be used to deter inattentive lane changing. Furthermore, one unit increase in longitudinal acceleration of the leading vehicle is associated with 0.1 % reduction but 0.8 % increase in the risk of slight and severe vehicle-vehicle conflicts. Also, one unit increase in longitudinal acceleration of the following vehicle is associated with 0.4 % reduction in the risk of severe vehicle-vehicle conflict. However, one unit increase in vehicle longitudinal acceleration is associated with 2.3 % and 2.6 % increase in the risk of slight and severe vehicle-motorcycle conflicts. Such findings are consistent with that of previous studies at highway merging and diverging areas for aggressive driving behaviour (Wei et al., 2019; Stipancic et al., 2019; Song et al., 2022a; Ali et al., 2022). This justifies that speed change of vehicle can be considered as effective surrogate safety measure for both vehicle-vehicle and vehicle-motorcycle collisions at the intersections (Tageldin et al., 2017). Last but not least, factors that affect the means of the random parameters for vehicle-vehicle conflicts are identified. For the likelihood of slight conflict, favorable effect of longitudinal acceleration of leading vehicle is offset for rear-end conflict. This could be attributed to the compensatory strategy of risk-taking drivers in seemingly favorable traffic scenarios (Chen et al., 2021a).

For the motorcycle attributes, one unit increase in motorcycle lateral speed is associated with 0.08 % reduction in the risk of severe vehicle-motorcycle conflicts. This justifies the role of motorcycle's weaving movement in the risk reduction of vehicle-motorcycle collision. Indeed, lateral maneuvers such as lane changing and swerving are prevalent in pre-crash conditions (Liu et al., 2022; Nasernejad et al., 2023; Zaki et al., 2014). They may be indicative to risk of potential traffic conflict (Tageldin and Sayed, 2019). Additionally, factors that affect the means of the random parameters are also identified. For the likelihood of severe vehicle-motorcyclist conflict, random parameter of lateral speed of motorcyclist is positively correlated to that of the lateral speed of vehicle. Lateral speed is closely correlated to the prevalence of evasive actions like swerving. This can be considered as an effective surrogate safety measure (Lanzaro et al., 2022). Last but not least, this can be attributed to the compensatory strategy adopted by the risk-averse drivers (Chen et al., 2021a).

6.2. Conflict type and signal time plan

For the conflict type, rear-end conflict is associated with 6.8 % increase in the risk of severe vehicle-vehicle conflicts and 9.4 % increase in that of severe vehicle-motorcycle conflicts. This justifies the role of yellow dilemma in the collision risk at the intersection approaches (Zhao et al., 2023). To mitigate the risk attributed to yellow dilemma, advanced traffic control measures like adaptive traffic signal time plan

Table 7
Marginal effects on the probabilities of vehicle-vehicle and vehicle-motorcycle conflicts.

Variable	Vehicle-vehicle conflict			Vehicle-motorcycle conflict		
	No conflict	Slight conflict	Severe conflict	No conflict	Slight conflict	Severe conflict
Longitudinal speed of (leading) vehicle [SL]	−0.068	0.080	−0.012	−0.032	0.054	−0.022
Longitudinal speed of (leading) vehicle [SE]	−0.029	−0.020	0.048	−0.020	−0.050	0.070
Longitudinal speed of following vehicle [SL]	−0.057	0.066	−0.010	NA	NA	NA
Longitudinal speed of following vehicle [SE]	−0.025	−0.018	0.043	NA	NA	NA
Lateral speed of (leading) vehicle [SL]	−0.020	0.024	−0.004	−0.016	0.030	−0.014
Lateral speed of (leading) vehicle [SE]	−0.011	−0.008	0.019	−0.007	−0.019	0.026
Longitudinal acceleration of (leading) vehicle [SL]	0.001	−0.001	0.000	−0.012	0.023	−0.011
Longitudinal acceleration of (leading) vehicle [SE]	−0.007	−0.001	0.008	−0.006	−0.020	0.026
Longitudinal acceleration of following vehicle [SL]	IS	IS	IS	NA	NA	NA
Longitudinal acceleration of following vehicle [SE]	0.003	0.002	−0.004	NA	NA	NA
Motorcycle lateral speed [SL]	NA	NA	NA	IS	IS	IS
Motorcycle lateral speed [SE]	NA	NA	NA	0.0006	0.0003	−0.0008
Rear-end conflict [SL]	IS	IS	IS	IS	IS	IS
Rear-end conflict [SE]	−0.043	−0.025	0.068	−0.031	−0.062	0.094
Time to green [SL]	IS	IS	IS	IS	IS	IS
Time to green [SE]	IS	IS	IS	−0.013	−0.026	0.039

SL – Slight conflict; SE – Severe conflict; IS – Insignificant; **bold** – direct average marginal effects; NA – Not included in the model.

can be implemented (Jin et al., 2021). Last but not least, one unit increase in time to green is associated with 3.9 % increase in the risk of severe vehicle-motorcycle conflicts. This could be attributed to the anxiety of road users, especially motorcyclists tend to be more risk taking and aggressive (Song et al., 2022a). This is indicative to the effectiveness of optimal signal time allocation and signal countdown (to green) device in mitigating the potential collision risk attributed to driver anxiety (Islam et al., 2017).

6.3. Summary

Findings and implications of this study are summarized in Table 8. As shown in Table 8, findings are generally similar between vehicle-vehicle and vehicle-motorcycle conflicts. For example, both vehicle longitudinal speed and lateral speed are positively associated with the risk of vehicle-vehicle and vehicle-motorcycle conflicts. Additionally, rear-end conflict has a higher likelihood of being severe for both vehicle-vehicle and vehicle-motorcycle conflicts. In contrast, vehicle longitudinal acceleration is positively associated with the risk of both vehicle-vehicle and vehicle-motorcycle conflicts, but negatively associated with that of

Table 8
Comparison between vehicle-vehicle and vehicle-motorcycle conflicts.

Variable	Vehicle-vehicle conflict		Vehicle-motorcycle conflict		Implications
	Slight conflict	Severe conflict	Slight conflict	Severe conflict	
Longitudinal speed of (leading) vehicle	+	+	+	+	Speed reduction using warning signs and rumble strips, and traffic control measures like automated speed enforcement camera
Longitudinal speed of following vehicle	+	+	NA	NA	Remedial measures like improved pavement surface and clearer road markings
Lateral speed of (leading) vehicle	+	+	+	+	Reduction of driver anxiety using advanced traffic management like adaptive signal time plan and signal countdown (to red) device
Longitudinal acceleration of (leading) vehicle	− [^]	+	+	+	Remedial measures like improved pavement surface and clearer road markings
Longitudinal acceleration of following vehicle	IS	−	NA	NA	Reduction of driver anxiety using advanced traffic management like adaptive signal time plan and signal countdown (to green) device
Motorcycle lateral speed	NA	NA	IS	− [^]	Elimination of yellow dilemma using adaptive signal time plan
Time to green	IS	IS	IS	+	
Rear-end conflict	IS	+	IS	+	

+ Positive association; − Negative association; NA – Not used variable; IS – Insignificant; [^]With heterogeneity in means and variances.

slight vehicle-vehicle conflict. Last but not least, time-to-green is significantly associated with the risk of severe vehicle-motorcycle conflict only.

Above findings are indicative to the advanced traffic management measures in reducing the speed of vehicles approaching the intersections (Guo et al., 2016). For example, automated speed enforcement cameras and rectangular Rapid-Flashing Beacons (RRFB) can be installed at the accident-prone locations (Pezeshknejad and Rowangould, 2024). Additionally, rumble strips and chevron pavement markings can be used (Guo et al., 2016). Furthermore, drivers' anxiety in the yellow dilemma should be accommodated. This is indicative to the adaptive signal time plan (Islam et al., 2017; Chen et al., 2021a). Last but not least, the trade-off between safety and level of comfort of car occupancy should be considered when estimating the optimal speed and acceleration of vehicles. This is indicative to the motion planning strategy of automated driving system (Chen et al., 2021b).

7. Conclusions

Safety of signalized intersections has been of major concern. Influencing factors including road environment, vehicle type, and road user characteristics that affect the crash risk and propensity of dangerous behaviour like speeding and red-light running are identified. However, safety of vulnerable road user likes motorcycle is less studied. It is rare that vehicle dynamics and their effects on potential collision risk are considered. In this study, the multinomial logit regression approach is adopted to examine the relationship between possible factors and the risk of traffic conflict involving and not involving motorcycle at the signalized intersection, based on the high-definition trajectory data obtained from drone video. In addition to signal time plan and conflict type, individual vehicle and motorcycle attributes like longitudinal speed, lateral speed and longitudinal acceleration are also considered. Furthermore, the effect of unobserved heterogeneity is accounted using random parameters model with heterogeneity in means and variances.

Results indicate that risk of vehicle-vehicle and vehicle-motorcycle conflicts are positively associated with vehicle longitudinal speed, lateral speed, and longitudinal acceleration, except that risk of slight vehicle-vehicle conflict and longitudinal acceleration are negatively correlated. This may be attributed to the association between the risk of severe conflict and variation in traffic speed (Song et al., 2022a; Nasernejad et al., 2023). Additionally, likelihoods of severe vehicle-vehicle and vehicle-motorcycle conflicts are higher for rear-end conflict. This could be attributed to the yellow dilemma at the intersection approaches. Furthermore, the likelihood of severe vehicle-motorcycle conflict would increase with time to green. However, the likelihood of severe vehicle-motorcycle conflict would decrease with motorcycle longitudinal speed. Such findings are indicative to the remedial measures like advanced warning signs, rumble strips and chevron pavement markings, automated speed enforcement camera, and signal countdown device, especially at the accident-prone locations where motorcycles are prevalent (Chen et al., 2020).

To generalize the policy applications of this study, it is worth investigating the conflict risk involving all vulnerable road user types including pedestrians and bicycles, especially in the urban areas (Bai and Sze, 2020; Zhu and Sze, 2021; Zhu et al., 2021). Additionally, it is worth exploring the effect of unobserved heterogeneity attributed to driving style, aggressive, and compensatory strategy of individual drivers on the conflict risk, using advanced analytic methods like latent structure, deep learning, and fuzzy logic approaches (Song et al., 2022a). Last but not least, effects of spatial dependency and temporal instability should be accounted for when more comprehensive data at multiple sites are available in the future study (Zeng et al., 2019). Nevertheless, conflicts involving non-motorized transport modes including pedestrians and bicycles are not considered in this study. In the future study, it is worth exploring the interferences between conflicts involving and not involving non-motorized transport modes.

CRediT authorship contribution statement

Shile Zhang: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **N.N. Sze:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- Abdel-Aty, M., Wang, X., 2006. Crash estimation at signalized intersections along corridors: analyzing spatial effect and identifying significant factors. *Transp. Res. Rec.* 1953, 98–111.
- Ali, Y., Bliemer, M.C., Haque, M.M., Zheng, Z., 2022. Examining braking behaviour during failed lane-changing attempts in a simulated connected environment with driving aids. *Transp. Res. C* 136, 103531.
- Alnawmasi, N., Mannering, F., 2022. The impact of higher speed limits on the frequency and severity of freeway crashes: Accounting for temporal shifts and unobserved heterogeneity. *Anal. Methods Accid. Res.* 34, 100205.
- Arun, A., Haque, M.M., Washington, S., Sayed, T., Mannering, F., 2021. A systematic review of traffic conflict-based safety measures with a focus on application context. *Anal. Methods Accid. Res.* 32, 100185.
- Bai, L., Sze, N.N., 2020. Red light running behavior of bicyclists in urban area: Effects of bicycle type and bicycle group size. *Travel Behav. Soc.* 21, 226–234.
- Bao, J., Liu, P., Ukkusuri, S.V., 2019. A spatiotemporal deep learning approach for citywide short-term crash risk prediction with multi-source data. *Accid. Anal. Prev.* 122, 239–254.
- Behnood, A., Mannering, F., 2017a. The effect of passengers on driver-injury severities in single-vehicle crashes: A random parameters heterogeneity-in-means approach. *Anal. Methods Accid. Res.* 14, 41–53.
- Behnood, A., Mannering, F., 2017b. Determinants of bicyclist injury severities in bicycle-vehicle crashes: A random parameters approach with heterogeneity in means and variances. *Anal. Methods Accid. Res.* 16, 35–47.
- Chang, F., Yasmin, S., Huang, H., Chan, A.H.S., Haque, M.M., 2021. Injury severity analysis of motorcycle crashes: A comparison of latent class clustering and latent segmentation based models with unobserved heterogeneity. *Anal. Methods Accid. Res.* 32, 100188.
- Chen, A.Y., Chiu, Y.L., Hsieh, M.H., Lin, P.W., Angah, O., 2020a. Conflict analytics through the vehicle safety space in mixed traffic flows using UAV image sequences. *Transp. Res. C* 119, 102744.
- Chen, T., Sze, N.N., Saxena, S., Pinjari, A.R., Bhat, C.R., Bai, L., 2020b. Evaluation of penalty and enforcement strategies to combat speeding offences among professional drivers: A Hong Kong stated preference experiment. *Accid. Anal. Prev.* 135, 105366.
- Chen, T., Sze, N.N., Newnam, S., Bai, L., 2021a. Effectiveness of the compensatory strategy adopted by older drivers: Difference between professional and non-professional drivers. *Transp. Res. F* 77, 168–180.
- Chen, T., Wong, Y.D., Shi, X., Yang, Y., 2021b. A data-driven feature learning approach based on Copula-Bayesian Network and its application in comparative investigation on risky lane-changing and car-following maneuvers. *Accid. Anal. Prev.* 154, 106061.
- Chen, T., Lu, Y., Fu, X., Sze, N.N., Ding, H., 2022. A resampling approach to disaggregate analysis of bus-involved crashes using panel data with excessive zeros. *Accid. Anal. Prev.* 164, 106496.
- Cooper, P.J., 1984. Experience with traffic conflicts in Canada with emphasis on “post encroachment time” techniques. *Int. Calib. Study Traffic Conflict Tech.* 75–96.
- Cooper, D.F., Ferguson, N., 1976. Traffic studies at T-Junctions. 2. A conflict simulation Record. *Traffic Eng. Control* 17.
- Das, A., Khan, M.N., Ahmed, M.M., 2020. Detecting lane change maneuvers using SHRP2 naturalistic driving data: A comparative study machine learning techniques. *Accid. Anal. Prev.* 142, 105578.
- Ding, H., Lu, Y., Sze, N.N., Chen, T., Guo, Y., Lin, Q., 2022. A deep generative approach for crash frequency model with heterogeneous imbalanced data. *Anal. Methods Accid. Res.* 34, 100212.
- Essa, M., Sayed, T., 2018. Traffic conflict models to evaluate the safety of signalized intersections at the cycle level. *Transp. Res. C* 89, 289–302.
- Figuroa-Medina, A.M., 2005. Subjective and objective risks consideration in modeling highway safety. Purdue University.
- Fountas, G., Sarwar, M.T., Anastasopoulos, P.C., Blatt, A., Majka, K., 2018. Analysis of stationary and dynamic factors affecting highway accident occurrence: A dynamic correlated grouped random parameters binary logit approach. *Accid. Anal. Prev.* 113, 330–340.
- Fu, T., Miranda-Moreno, L., Saunier, N., 2018. A novel framework to evaluate pedestrian safety at non-signalized locations. *Accid. Anal. Prev.* 111, 23–33.
- Gao, D., Zhang, X., 2024. Injury severity analysis of single-vehicle and two-vehicle crashes with electric scooters: A random parameters approach with heterogeneity in means and variances. *Accid. Anal. Prev.* 195, 107408.
- Goswamy, A., Abdel-Aty, M., Islam, Z., 2023. Factors affecting injury severity at pedestrian crossing locations with Rectangular RAPID Flashing Beacons (RRFB) using XGBoost and random parameters discrete outcome models. *Accid. Anal. Prev.* 181, 106937.
- Guo, Y., Liu, P., Liang, Q., Wang, W., 2016. Effects of parallelogram-shaped pavement markings on vehicle speed and safety of pedestrian crosswalks on urban roads in China. *Accid. Anal. Prev.* 95, 438–447.
- Guo, Y., Essa, M., Sayed, T., Haque, M.M., Washington, S., 2019. A comparison between simulated and field-measured conflicts for safety assessment of signalized intersections in Australia. *Transp. Res. C* 101, 96–110.
- Hayward, J.C., 1972. Near miss determination through use of a scale of danger. *Highw. Res. Rec.* 384, 22–34.
- Huang, H., Chin, H.C., Haque, M.M., 2008. Severity of driver injury and vehicle damage in traffic crashes at intersections: A Bayesian hierarchical analysis. *Accid. Anal. Prev.* 40, 45–54.
- Hydén, C., 1987. The development of a method for traffic safety evaluation: The Swedish Traffic Conflicts Technique. Lund University. PhD Thesis.
- Islam, Z., Abdel-Aty, M., 2023. Traffic conflict prediction using connected vehicle data. *Anal. Methods Accid. Res.* 39, 100275.
- Islam, M.R., Wyman, A.A., Hurwitz, D.S., 2017. Safer driver responses at intersections with green signal countdown timers. *Transp. Res. F* 51, 1–13.
- Jin, W., Chowdhury, M., Salek, M.S., Khan, S.M., Gerard, P., 2021. Investigating hierarchical effects of adaptive signal control system on crash severity using random-parameter ordered regression models incorporating observed heterogeneity. *Accid. Anal. Prev.* 150, 105895.
- Jin, J., Huang, H., Yuan, C., Li, Y., Zou, G., Xue, H., 2023. Real-time crash risk prediction in freeway tunnels considering features interaction and unobserved heterogeneity: A two-stage deep learning modeling framework. *Anal. Methods Accid. Res.* 1, 100306.
- Khattak, Z.H., Magalotti, M.J., Fontaine, M.D., 2018. Estimating safety effects of adaptive signal control technology using the Empirical Bayes method. *Journal of Safety Research* 64, 121–128.
- Lanzaro, G., Sayed, T., Alsaleh, R., 2022. Can motorcyclist behavior in traffic conflicts be modeled? A deep reinforcement learning approach for motorcycle-pedestrian interactions. *Transportmetrica B* 10, 396–420.
- Lee, J., Abdel-Aty, M., Choi, K., Huang, H., 2015. Multi-level hot zone identification for pedestrian safety. *Accid. Anal. Prev.* 76, 64–73.
- Li, Y., Gu, R., Lee, J., Yang, M., Chen, Q., Zhang, Y., 2021b. The dynamic tradeoff between safety and efficiency in discretionary lane-changing behavior: A random parameters logit approach with heterogeneity in means and variances. *Accid. Anal. Prev.* 153, 106036.
- Li, H., Zhu, M., Graham, D.J., Ren, G., 2021a. Evaluating the speed camera sites selection criteria in the UK. *J. Saf. Res.* 76, 90–100.
- Lin, W., Wei, H., 2023. CAV-enabled data analytics for enhancing adaptive signal control safety environment. *Accid. Anal. Prev.* 192, 107290.
- Liu, Y., Alsaleh, R., Sayed, T., 2022. Modeling lateral interactions between motorized vehicles and non-motorized vehicles in mixed traffic using accelerated failure duration model. *Transportmetrica A* 18, 910–933.
- Ma, Z., Lu, X., Chien, S.I.J., Hu, D., 2018. Investigating factors influencing pedestrian injury severity at intersections. *Traffic Inj. Prev.* 19, 159–164.
- Nasernejad, P., Sayed, T., Alsaleh, R., 2023. Multiagent modeling of pedestrian-vehicle conflicts using adversarial inverse reinforcement learning. *Transportmetrica A* 19, 2061081.
- National Bureau of Statistics of China. China Statistical Yearbook. <http://www.stats.gov.cn> (accessed on 2024.03.20).
- Ospina-Mateus, H., Quintana Jiménez, L., López-Valdés, F.J., 2023. Analyzing traffic conflicts and the behavior of motorcyclists at unsignalized three-legged and four-legged intersections in Cartagena, Colombia. *Accid. Anal. Prev.* 191, 107222.
- Patel, D., Hosseini, P., Jalayer, M., 2023. A framework for proactive safety evaluation of intersection using surrogate safety measures and non-compliance behavior. *Accid. Anal. Prev.* 192, 107264.
- Pezeshknejad, P., Rowangould, D., 2024. Evaluating safety and compliance of pedestrian crossings in rural contexts: A before and after study of RRFBs and LED-embedded signs. *Accid. Anal. Prev.* 198, 107462.
- Seraneeprakarn, P., Huang, S., Shankar, V., Mannering, F., Venkataraman, N., Milton, J., 2017. Occupant injury severities in hybrid-vehicle involved crashes: A random parameters approach with heterogeneity in means and variances. *Anal. Methods Accid. Res.* 15, 41–55.
- Song, Y., Chitturi, M.V., Bremer, W.F., Bill, A.R., Noyce, D.A., 2022b. Review of United States research and guidelines on left turn lane offset: unsignalized intersections and signalized intersections with permitted left turns. *J. Traffic Transp. Eng. (English Edition)* 9 (4), 556–570.
- Song, L., Li, Y., Fan, D., Wu, P., 2020. Modeling pedestrian-injury severities in pedestrian-vehicle crashes considering spatiotemporal patterns: Insights from different hierarchical Bayesian random-effects models. *Anal. Methods Accid. Res.* 28, 100137.

- Song, P., Sze, N.N., Zheng, O., Abdel-Aty, M., 2022a. Addressing unobserved heterogeneity at road user level for the analysis of conflict risk at tunnel toll plaza: A correlated grouped random parameters logit approach with heterogeneity in means. *Anal. Methods Accid. Res.* 36, 100243.
- Stipanovic, J., Miranda-Moreno, L., Saunier, N., 2018. Vehicle maneuvers as surrogate safety measures: Extracting data from the GPS-enabled smartphones of regular drivers. *Accid. Anal. Prev.* 115, 160–169.
- Stipanovic, J., Miranda-Moreno, L., Saunier, N., Labbe, A., 2019. Network screening for large urban road networks: Using GPS data and surrogate measures to model crash frequency and severity. *Accid. Anal. Prev.* 125, 290–301.
- Sun, J., Qi, X., Xu, Y., Tian, Y., 2020. Vehicle turning behavior modeling at conflicting areas of mixed-flow intersections based on deep learning. *IEEE Trans. Intell. Transp. Syst.* 21 (9), 3674–3685.
- Sun, Z., Wang, D., Gu, X., Abdel-Aty, M., Xing, Y., Wang, J., Lu, H., Chen, Y., 2023. A hybrid approach of random forest and random parameters logit model of injury severity modeling of vulnerable road users involved crashes. *Accid. Anal. Prev.* 192, 107235.
- Sze, N.N., Su, J., Bai, L., 2019. Exposure to pedestrian crash based on household survey data: Effect of trip purpose. *Accid. Anal. Prev.* 128, 17–24.
- Tageldin, A., Sayed, T., 2019. Models to evaluate the severity of pedestrian-vehicle conflicts in five cities. *Transportmetrica A* 15, 354–375.
- Tageldin, A., Sayed, T., Shaaban, K., 2017. Comparison of time-proximity and evasive action conflict measures: Case studies from five cities. *Transp. Res. Rec.* 2661, 19–29.
- Tarko, A.P., 2012. Use of crash surrogates and exceedance statistics to estimate road safety. *Accid. Anal. Prev.* 45, 230–240.
- Tarko, A.P., 2023. Maximum likelihood method of estimating the conflict-crash relationship. *Accid. Anal. Prev.* 179, 106875.
- Tarko, A.P., Davis, G., Saunier, N., Sayed, T., Washington, S., 2009. White paper: surrogate measures of safety. Committee on Safety Data Evaluation and Analysis (ANB20).
- Tarko, A.P., Lizarazo, C.G., 2021. Validity of failure-caused traffic conflicts as surrogates of rear-end collisions in naturalistic driving studies. *Accid. Anal. Prev.* 149, 105863.
- Thomas, L.J., Srinivasan, R., Decina, L.E., Staplin, L., 2008. Safety effects of automated speed enforcement programs: critical review of international literature. *Transp. Res. Rec.* 2078, 117–126.
- Tsui, K.L., So, F.L., Sze, N.N., Wong, S.C., Leung, T.F., 2009. Misclassification of injury severity among road casualties in police reports. *Accid. Anal. Prev.* 41, 84–89.
- Wang, L., Abdel-Aty, M., Lee, J., 2017. Safety analytics for integrating crash frequency and real-time risk modeling for expressways. *Accid. Anal. Prev.* 104, 58–64.
- Wang, L., Abdel-Aty, M., Ma, W., Hu, J., Zhong, H., 2019b. Quasi-vehicle-trajectory-based real-time safety analysis for expressways. *Transp. Res. C* 103, 30–38.
- Wang, K., Bhowmik, T., Yasmin, S., Zhao, S., Eluru, N., Jackson, E., 2019a. Multivariate copula temporal modeling of intersection crash consequence metrics: A joint estimation of injury severity, crash type, vehicle damage and driver error. *Accid. Anal. Prev.* 125, 188–197.
- Wang, X., Yuan, J., Schultz, G., Fang, S., 2018. Investigating the safety impact of roadway network features of suburban arterials in Shanghai. *Accid. Anal. Prev.* 113, 137–148.
- Washington, S., Karlaftis, M., Mannering, F., 2011. *Statistical and Econometric Methods for Transportation Data Analysis*, second ed. Chapman and Hall/CRC, Boca Raton, FL.
- Watson, A., Watson, B., Vallmuur, K., 2015. Estimating under-reporting of road crash injuries to police using multiple linked data collections. *Accid. Anal. Prev.* 83, 18–25.
- Wei, Y., Li, K., Tang, K., 2019. Trajectory-based identification of critical instantaneous decision events at mixed-flow signalized intersections. *Accid. Anal. Prev.* 123, 324–335.
- Wong, S.C., Sze, N.N., Li, Y.C., 2007. Contributory factors to traffic crashes at signalized intersections in Hong Kong. *Accid. Anal. Prev.* 39, 1107–1113.
- World Health Organization. Road Traffic Injuries. <https://www.who.int/health-topics/road-safety> (accessed on 20 March 2024).
- Wu, Y., Abdel-Aty, M., Zheng, O., Cai, Q., Zhang, S., 2020. Automated safety diagnosis based on unmanned aerial vehicle video and deep learning algorithm. *Transp. Res. Rec.* 2674 (8), 350–359.
- Wu, J., Xu, H., Zheng, Y., Tian, Z., 2018. A novel method of vehicle-pedestrian near-crash identification with roadside LiDAR data. *Accid. Anal. Prev.* 121, 238–249.
- Xie, K., Wang, X., Huang, H., Chen, X., 2013. Corridor-level signalized intersection safety analysis in Shanghai, China using Bayesian hierarchical models. *Accid. Anal. Prev.* 50, 25–33.
- Xie, K., Di, Y., Kaan, O., Hong, Y., 2019. Use of real-world connected vehicle data in identifying high-risk locations based on a new surrogate safety measure. *Accid. Anal. Prev.* 125, 311–319.
- Xin, C., Guo, R., Wang, Z., Lu, Q., Lin, P.S., 2017. The effects of neighborhood characteristics and the built environment on pedestrian injury severity: A random parameters generalized ordered probability model with heterogeneity in means and variances. *Anal. Methods Accid. Res.* 16, 117–132.
- Xu, Y., Shao, W., Li, J., Yang, K., Wang, W., Huang, H., Lv, C., Wang, H., 2022. SIND: A drone dataset at signalized intersection in China. *IEEE 25th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE.
- Yan, W., Wong, S.C., Loo, B.P.Y., Wu, C.Y.H., Huang, H., Pei, X., Meng, F., 2022. An assessment of the effect of green signal countdown timers on drivers' behavior and on road safety at intersections, based on driving simulator experiments and naturalistic observation studies. *J. Saf. Res.* 82, 1–12.
- Yu, M., Zheng, C., Ma, C., 2020. Analysis of injury severity of rear-end crashes in work zones: A random parameters approach with heterogeneity in means and variances. *Anal. Methods Accid. Res.* 27, 100126.
- Yuan, J., Abdel-Aty, M., 2018. Approach-level real-time crash risk analysis for signalized intersections. *Accid. Anal. Prev.* 119, 274–289.
- Zaki, M.H., Sayed, T., Shaaban, K., 2014. Use of drivers' jerk profiles in computer vision-based traffic safety evaluations. *Transp. Res. Rec.* 2434 (1), 103–112.
- Zeng, Q., Gu, W., Zhang, X., Wen, H., Lee, J., Hao, W., 2019. Analyzing freeway crash severity using a Bayesian spatial generalized ordered logit model with conditional autoregressive priors. *Accid. Anal. Prev.* 127, 87–95.
- Zhai, X., Huang, H., Sze, N.N., Song, Z., Hon, K.K., 2019. Diagnostic analysis of the effects of weather condition on pedestrian crash severity. *Accid. Anal. Prev.* 122, 318–324.
- Zhang, S., Abdel-Aty, M., 2022. Real-Time crash potential prediction on freeways using connected vehicle data. *Anal. Methods Accid. Res.* 36, 100239.
- Zhang, S., Abdel-Aty, M., Wu, Y., Zheng, O., 2022a. Pedestrian crossing intention prediction at red-light using pose estimation. *IEEE Trans. Intell. Transp. Syst.* 23 (3), 2331–2339.
- Zhang, Y., Li, H., Ren, G., 2022b. Estimating heterogeneous treatment effects in road safety analysis using generalized random forests. *Accid. Anal. Prev.* 165, 106507.
- Zhao, J., Yu, J., Zhang, F., Liu, Y., 2023. Mitigation of signalized intersection collision risks with trajectory based dynamic dilemma zone protection. *Accid. Anal. Prev.* 192, 107288.
- Zheng, L., Sayed, T., Essa, M., 2019. Validating the bivariate extreme value modeling approach for road safety estimation with different traffic conflict indicators. *Accid. Anal. Prev.* 123, 314–323.
- Zheng, L., Sayed, T., Mannering, F., 2021. Modeling traffic conflicts for use in road safety analysis: A review of analytic methods and future directions. *Anal. Methods Accid. Res.* 29, 100142.
- Zhu, D., Sze, N.N., 2021. Propensities of red light running of pedestrians at the two-stage crossings with split pedestrian signal phases. *Accid. Anal. Prev.* 151, 105958.
- Zhu, D., Sze, N.N., Bai, L., 2021. Roles of personal and environmental factors in the red light running propensity of pedestrian: Case study at the urban crosswalks. *Transp. Res. F* 76, 47–58.
- Zhu, D., Sze, N.N., Feng, Z., Yang, Z., 2022a. A two-stage safety evaluation model for the red-light running behaviour of pedestrians using the game theory. *Saf. Sci.* 147, 105600.
- Zhu, M., Sze, N.N., Newnam, S., 2022b. Effect of urban street trees on pedestrian safety: A micro-level pedestrian casualty model using multivariate Bayesian spatial approach. *Accid. Anal. Prev.* 176, 106818.